DMSD Assignment-5

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* Consider a relation R(ABCDEFGHIJ) with the following set of functional dependencies G (~~C~~ → AB, CD → E, C → FG, H → IJ, D → H)

(1) Is CDE a superkey of R? (w.r.t G)?

A. Yes
since $CDE^+ = ABCDEFGHIJ = R$

(2) Is CDS a key of R? (w.r.t G)?

A. CD is a proper subset of CDE (CD ⊂ CDE)

CD is a superkey of R: $CD^+ = R$

since a proper superset of CDE is a superkey of R,

* CDE is not a key of R.

③ Apply the Appropriate algorithm a key for R? (w.r.t G)?

A. C and D do not appear in the RHs of any FD in F. Therefore, every key of R must contain C&D. We will not try to Remove them from K.

Initially K=R

If $(K-x)^+ = R$ for some attribute x in K-{CD} then we remove x from K; otherwise, we have x in K and continue with another attribute in

K-{CD}, until all attributes in K-{CD} are

we try to remove E from R: (CDFGHIJAB)

$t = R \ldots iK_i = CDFGHIJAB$

" " " F from R: (CDGHIJAB)+ = R.·.K: = CDGHIJAB

" " " G from R: (CDHIJAB)+ = R.·.K: = CDHIJAB

" " " H from R: (CDIJAB)+ = R.·.K: = CDIJAB

" " " I from R: (CDJAB)+ = R.·.K: = CDJAB

" " " J from R: (CDAB)+ = R.·.K: = CDAB

" " " A from R: (CDB)+ R.·.K: = CDB

" " " B from R: (CD)+ = R.·.K: = CD

Thus; CD is A key of R.

(4) Apply the Appropriate algorithm to determine all the key for R? (w.r.t G)?

A. C & D do not appear in the RHS. of any FD in F. Therefore, every key of R must contain C & D from Question 2, CD is a superkey of R. Thus, CD is the unique key of R.

(5) Determine the Prime Attributes of R?

A. The Prime Attribute of R are C&D.

(6) Is R in BCNF? (w.r.t G)?

A. Consider H → IJ in F

H → IJ violates BCNF in R: H is not a super key of ($H^+ = HIJ \neq R$)

R is not in BCNF

(7) Is R in 3NF? (wrt G)?

A. Consider H → IJ in F.

H → IJ violates 3NF in R: H is not a super key of R ($H^+ = HIJ \neq R$) & neither I or J is a prime attribute of R.

(8) Determine whether the decomposition $D = \{CDE, CFG, DH, HIJ, FAB\}$ has

i) the dependency Presentation Property &

A. Yes

Since every FD in F is applicable to atleast one schema in D. For instance, $CD \supset E$

$\in \Pi_{CDE}(F)$.

(ii) the Lossless Join Property, with respect to G. Also determine which normal for each relation in the decomposition is in.

A. We use the property of binary LJ decomposition & the property of repeated LJ decomposition.

Let $R_1 = CDE$, $R_2 = CFG$ & $R_3 = R_1 \cup R_2 = CDEFG$

Since $G \models C \to FG$, $G \models R_1 \cap R_2 \to R_2 - R_1$.

Thus, the decomposition $\{R_1, R_2\}$ of R_3 is LJ w.r.t $\pi_{R_3}(G)$.

Let $R_4 = DH$ & $R_5 = R_3 \cup R_4 = CDEFGH$

Since $G \models D \to H$, $G \models R_3 \cap R_4 \to R_4 - R_3$.

Thus, the decomposition $\{R_3, R_4\}$ of R_5 is LJ w.r.t $\pi_{R_5}(F)$

Therefore, the decomposition $\{R_1, R_2, R_4\}$ of R_5 is LJ w.r.t $\pi_{R_5}(F)$.

Let $R_6 = HIJ$ and $R_7 = R_5 \cup R_6 = CDEFGHIJ$

Since $G \models H \to IJ$, $G \models R_5 \cap R_6 \to R_6 - R_5$

Thus, the decomposition $\{R_5, R_6\}$ of R_7 is LJ, w.r.t $\pi_{R_7}(G)$.

Therefore, the decomposition $\{R_1, R_2, R_4, R_7\}$ of R_7 is LJ w.r.t. $\pi_{R_7}(G)$.

Let $R_8 = FAB$. Notice that $R_7 \cup R_8 = R$.

Since $G1 = F \to AB$, $G1 = R_7 \cap R_8 \to R_8 - R_7$.

Thus, the decomposition (R_7, R_8) of R is LJ w.r.t. F.

* Therefore, the decomposition ~~(R_7, R_8) of R is LJ~~ $\{R_1, R_2, R_4, R_6, R_8\}$ of R is LJ w.r.t. G.

(iii) CDE: $\pi_{CDE}(G) = \{CD \to E\}$. Since CD is super key of CDE, CDE is in BCNF.

Similarly we show that the rest of the schemas in D are in BCNF.

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